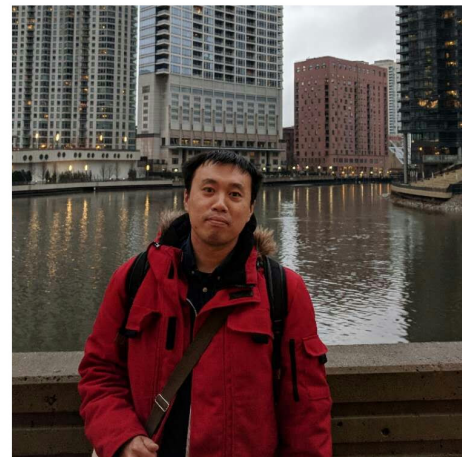


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Personal

Date on Birth: 27 April 1982.

Nationality: Japan.

Gender: Male.

Education

Ph.D(Mathematics). Tohoku University, September 2010.

M.S(Mathematics). Tohoku University, March 2007.

B.S(Mathematics). Saitama University, March 2005.

Employment

Associate professor at Department of Mathematics, College of Science and Technology, Nihon University, April 2015 – present.

Assistant professor at Department of Mathematics, College of Science and Technology, Nihon University, April 2012 – March 2015.

Post-doctoral researcher at Department of Mathematics, Graduates School of Science, Hokkaido University, April 2011 – March 2012.

Research Fellow of the Japan Society for the Promotion of Science, April 2009 – March 2011.

Research assistant of the global COE at Tohoku University, August 2008 – March 2009.

Research assistant of the Daigakuin GP at Tohoku University, January 2008 – May 2008.

Research assistant of the COE at Tohoku University, April 2007 – March 2008.

Research

Research interest: Nonlinear analysis, Regularity theory for nonlinear partial differential equations, geometric measure theory and its application, Mathematical modeling for grain boundary motion.

Journal articles with peer review

- Takashi Kagaya, Masashi Mizuno, Keisuke Takasao, *Long time behavior for a curvature flow of networks related to grain boundary motion with the effect of lattice misorientations*, *Ann. Sc. Norm. Super. Pisa Cl. Sci. (5)* **25** (2024), 2369–2450.
- Xianjin Chen, Chiun-Chang Lee, Masashi Mizuno, *Unified asymptotic analysis and numerical simulations of singularly perturbed linear differential equations under various nonlocal boundary effects*, *Commun. Math. Sci. Communications in Mathematical Sciences*. **22** (2024), 394–434.
- Yekaterina Epshteyn, Chang Liu, Chun Liu, Masashi Mizuno, *Local well-posedness of a nonlinear Fokker-Planck model*, *Nonlinearity* **36** (2023), 1890–1917.
- Chiun-Chang Lee, Masashi Mizuno, Sang-Hyuck Moon, *On the uniqueness of linear convection-diffusion equations in large domains with integral boundary conditions*, *C. R. Math. Acad. Sci. Paris* **361** (2023), 191–206.
- Yekaterina Epshteyn, Chang Liu, Chun Liu, Masashi Mizuno, *Nonlinear inhomogeneous Fokker-Planck models: energetic-variational structures and long time behavior*, *Anal. Appl. (Singap.)* **20** (2022), 1295–1356.
- Yekaterina Epshteyn, Chun Liu, Masashi Mizuno, *A stochastic model of grain boundary dynamics: A Fokker-Planck perspective*, *Math. Models Methods Appl. Sci.* **32** (2022), 2189–2236.
- Katayun Barmak, Anastasia Dunca, Yekaterina Epshteyn, Chun Liu, Masashi Mizuno, *Grain Growth and the Effect of Different Time Scales*, in “Research in Mathematics of Materials Science,” 33–58. *Assoc. Women Math. Ser.*, 31, Springer, Cham, 2022.
- Masashi Mizuno, Keisuke Takasao, *A curve shortening equation with time-dependent mobility related to grain boundary motions*, *Interfaces Free Bound.* **23** (2021), 169–190.
- Yekaterina Epshteyn, Chun Liu, Masashi Mizuno, *Large time asymptotic behavior of grain boundaries motion with dynamic lattice misorientations and with triple junctions drag*, *Commun. Math. Sci.* **19** (2021), 1403–1428.
- Yekaterina Epshteyn, Chun Liu, Masashi Mizuno, *Motion of grain boundaries with dynamic lattice misorientations and with triple junctions drag*, *SIAM J. Math. Anal.* **53** (2021), 3072–3097.
- Masashi Mizuno and Keisuke Takasao, *Gradient estimates for mean curvature flow with Neumann boundary conditions*, *NoDEA Nonlinear Differential Equations Appl.* **24** (2017), Art. 32, 24pp.
- Masashi Mizuno and Yoshihiro Tonegawa, *Erratum to "Convergence of the Allen-Cahn equation with Neumann boundary conditions"*, *SIAM J. Math. Anal.* **48** (2016), 3035–3036.
- Masashi Mizuno and Yoshihiro Tonegawa, *Convergence of the Allen-Cahn equation with Neumann boundary conditions*, *SIAM J. Math. Anal.* **47** (2015), 1906–1932.
- Masashi Mizuno and Takayoshi Ogawa, *Regularity and asymptotic behavior for the Keller-Segel system of degenerate type with critical nonlinearity*, *J. Math. Sci. Univ. Tokyo* **20** (2013), 375–433.
- Masashi Mizuno, *Hölder estimates for solutions of the Cauchy problem for the porous medium equation with external forces*, *Manuscripta Math.* **141** (2013), 273–313.
- Masashi Mizuno, *Remarks on Hölder continuity for solutions of the p -Laplace evolution equations*, *J. Math. Anal. Appl.* **382** (2011), 785–791.
- Masashi Mizuno, *Harnack estimates for some nonlinear parabolic equation*, *Differential and Integral Equations* **21** (2008), 693–716.

Preprints

Yekaterina Epshteyn, Chun Liu, and Masashi Mizuno, *Longtime Asymptotic Behavior of Nonlinear Fokker-Planck Type Equations with Periodic Boundary Conditions*.

Kouta Araki and Masashi Mizuno, *Long-time behavior of free energy in the nonlinear Fokker-Planck equation*.

Masashi Mizuno, Ayumi Sakiyama, and Keisuke Takasao, *The Łojasiewicz-Simon inequality related to grain boundary motion and its applications*.

Proceedings without peer review

Masashi Mizuno, *Mathematical modeling for grain boundary motion with dynamic lattice misorientations and triple junction drag* (Japanese), in Proceedings of 46th Sapporo Symposium on Partial Differential Equations, **181** (2021), 39–50.

Masashi Mizuno and Keisuke Takasao, *Gradient estimates for mean curvature flow with Neumann boundary conditions*, in RIMS Kôkyûroku, Theory of evolution equations and applications to nonlinear problems, **2066** (2018), 35–45.

Masashi Mizuno and Yoshihiro Tonegawa, *Convergence of the Allen-Cahn equation with Neumann boundary conditions*, in RIMS Kôkyûroku, Regularity and Singularity for Partial Differential Equations with Conservation Laws **1962** (2015), 10–16.

Masashi Mizuno and Takayoshi Ogawa, *Hölder continuity for some degenerate parabolic equation and its application*, in RIMS Kôkyûroku, Nonlinear evolution equations and mathematical modeling **1693** (2010), 45–56.

Grants(Principal Investigator)

Fund for the Promotion of Joint International Research (Fostering Joint International Research), JSPS KAKENHI Grant Number 25KK0052, 10,140,000 JPY, April 2026 – March 2029.

Grant-in-Aid for Scientific Research (C), JSPS KAKENHI Grant Number 22K03376, 4,030,000 JPY, April 2022 – March 2027.

Grant-in-Aid for Encouragement of Young Scientists, JSPS KAKENHI Grant Number 18K13446, 3,510,000 JPY, April 2018 – March 2022.

Grant-in-Aid for Encouragement of Young Scientists (B), JSPS KAKENHI Grant Number 25800084, 3,250,000 JPY, April 2014 – March 2016.

Grant-in-Aid for JSPS Fellows, JSPS KAKENHI Grant Number 09J01281, 1,400,000 JPY, April 2009 – March 2011.

Professional and society memberships

Member of Society for Industrial and Applied Mathematics, October 2019 – present.

Member of The Japan Society for Industrial and Applied Mathematics, April 2018 – present.

Member of Mathematical Society of Japan, October 2007 – present.

Professional activities

Organizing Seminars

Organizer of Research Analysis seminar, 1 February 2017 – present.

Organizer of Saitama Mathematical Analysis seminar, 1 April 2012 – present.

Organizer of Sandaigaku PDE seminar, 1 April 2012 – 31 January 2017.

Organizer of PDE seminar at Hokkaido University, 1 April 2011 – 31 March 2012.

Organizing International Workshops

Organizer of “The University of Newcastle-Nihon University joint workshop on Applied Mathematics,” 1 Nov 2025 – 4 Nov 2025.

Organizer of “Critical Phenomena in Nonlinear Partial Differential Equations, Harmonic Analysis, and Functional Inequalities,” 7 Nov 2023 – 10 Nov 2023.

Organizer of Mini-symposium “Mathematical Aspects of Multiscale Phenomena in Materials and Complex Fluids” in 10th International Congress on Industrial and Applied Mathematics, 23 August 2023.

Committee

Editorial board member of “Bulletin of the Japan Society for Industrial and Applied Mathematics,” April 2025 – March 2026.

A district representative of “Mathematical Society of Japan,” March 2025 – February 2026.

Vice editor-in-chief of “Bulletin of the Japan Society for Industrial and Applied Mathematics,” April 2023 – March 2025.

Editorial board member of “Bulletin of the Japan Society for Industrial and Applied Mathematics,” April 2020 – March 2023.

A district representative of “Mathematical Society of Japan,” March 2023 – February 2024.

A representative of “The Japan Society for Industrial and Applied Mathematics,” April 2023 – March 2026.